

FrameSeal

Review every frame before you share

Offline screen-recording redaction
intended for Snapdragon-powered HP PCs

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Proposal stage — implementation and hardware validation pending

Sensitive text can appear between clicks

The recording

A developer captures a browser, terminal or dashboard to explain a bug.

The exposure

An email or private identifier briefly appears during a window switch.

The review burden

The author must find the exposure and mask the correct frames before sharing.

Local analysis with human review

- 01 Import** Choose a short recording and email / exact-string rules.
- 02 Analyze** Run local OCR across every decoded frame.
- 03 Review** Inspect the timeline; adjust or add opaque masks.
- 04 Export** Write a separate silent video; block incomplete processing.

Processing coverage is not a guarantee that every secret was found.

On-device architecture

React UI + local FastAPI



CPU: decode + preprocess



EasyOCR on QNN HTP / Hexagon NPU



CPU: text decoding, policies + review



Complete + reviewed → masked export

Optional development cloud: AI Hub Workbench with synthetic data only.

A verified model route; a measurable NPU case

Models

Qualcomm EasyOCR detector + recognizer; checkpoint easyocr-small-stage1.

Deployment evidence

Qualcomm provides ONNX assets and a Windows-on-Snapdragon OCR example.

What we will verify

NPU graph placement, text recall and processing time versus CPU on the same HP PC.

Speed and energy gains remain hypotheses until measured.

A focused first version

MVP

- 720p clips up to 30 seconds
- English text
- Email and exact-string rules
- Every-frame analysis
- Review and silent export

Product constraints

- Offline after setup
- No upload of source footage
- Keyboard-accessible review
- Original file preserved
- No live meeting integration

No new model training or generative-AI service is required.

Evidence the demo must earn

Redaction quality

Missed sensitive regions; unnecessary masked area

Temporal coverage

One-, two- and five-frame exposures; exported-frame checks

Snapdragon execution

Provider logs and QNN profiling; fallback disabled

Performance + privacy

Same-device CPU/NPU time, memory and offline workflow

Results are pending. No benchmark numbers are asserted.

Delivery plan and current status

Prepared for the proposal

Problem, scoped workflow, model evidence, architecture and evaluation plan.

Implementation sequence

CPU OCR → video export → review UI → HP Snapdragon validation.

Judge demonstration

Find a brief synthetic exposure offline, review its mask and verify the saved video.

Application implementation and hardware validation are pending.